Self-Supervised Transfer Learning for Label-Efficient Multigas Electronic Nose Calibration

Dinh-Van Nguyen, DangThiThu Ha, Thu-Huong Nguyen , and NguyenDuc Hoa

Abstract*— Self-supervised transfer learning can reduce the number of labeled examples needed to calibrate a multigas electronic nose, although its benefits vary across gases and experimental conditions. We investigate this approach by first training a shared temporal encoder to reconstruct masked sensor responses without concentration labels, then fine-tuning it on a small set of labeled concentration pairs. Across ten random seeds, transfer learning reduces mean absolute error relative to training from scratch by 13.9% for NH3, 21.5% for H2, 27.5% for CO, and 25.7% for ethylene. These results come from six held-out identifier partitions for NH3/H2 using a predefined early-response window and three rolling chronological folds for CO/ethylene. To express label efficiency more directly, we also compare models trained with different calibration budgets. Using 10 labeled pairs, including validation labels, fine-tuning achieves approximately the NH3 and ethylene errors of scratch training with 20 pairs and a lower mean CO error, but does not match its H2 performance. Sensor identity and cross-sensor attention contribute unevenly, with NH3 showing the weakest evidence of benefit. Transfer also improves concentration regression in two of three future drift partitions when gas identity is known. Overall, the findings support selective use of self-supervised transfer for label-efficient calibration within the same sensor system. However, the proposed model does not outperform the strongest full-budget baseline on any target, and the NH3/H2 evaluation does not establish generalization across acquisition days.

***Index Terms*— Electronic nose, self-supervised learning, transfer learning, gas concentration regression, sensor drift, label-efficient calibration, sensor attention.**

I. Introduction

Electronic noses estimate gas properties from partially selective sensor arrays. Precise concentration labels require controlled delivery or reference measurement, making calibration costly across gases, mixtures and sessions. Sensor aging can require repeated calibration.

We seek to learn from inexpensive mixture-response recordings and reserve precise concentration measurements for a small calibration set. Preparation settings do not substitute for measured gas-phase ppm. Controlled-exposure datasets with labels withheld during pretraining test this premise retrospectively; they do not establish the cost or performance of a prospectively acquired unquantified mixture pool.

Limited-label electronic-nose research includes pseudo-labeling for recognition [1], [2] and domain adaptation, few-shot learning and meta-learning for drift [3]–[7]. Quantitative calibration remains less studied.

Temporal modeling [8]–[10], small-data transfer [11] and domain-adaptive regression [12] motivate reusable representations for concentration estimation.

Our shared temporal encoder learns response dynamics, while sensor identities and cross-sensor attention retain array structure. Masked intervals and channels are reconstructed before frozen or fine-tuned concentration regression. Two raw-response datasets support label-budget and multi-fold evaluation; a descriptor benchmark tests oracle-class regression under future drift. Generic SSL, chemometric, calibration-design and ablation controls retain adverse and null results. Figure 1 summarizes the pipeline.

II. Related Work

Fonollosa et al. [8] used reservoir computing to recover rapidly changing concentrations from slow sensor responses, demonstrating transient information beyond steady-state features.

Quantitative models include GAF-MTF/CNN-LSTM feature fusion [9], a lightweight transferred network with a single MEMS sensor [10], small-data parameter-feature transfer [11], and teacher-student domain-adaptive regression [12].

M-Training [1] and M-S4VMs [2] use predicted labels for gas recognition, with potential reinforcement of incorrect pseudo-labels.

Self-supervision instead derives targets from measurements. Time-series objectives include reconstruction, masking, contrastive and adversarial learning [13]. Masking intervals and channels encourages temporal and cross-sensor inference without concentration pseudo-labels.

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Drift changes the response distribution. TDACNN [3] and SAD-CNN [5] address domain differences; MDCN [4], CEGNN [6] and meta-learning [7] use limited reference data or transferable representations. Most 36-month-benchmark studies emphasize gas identity. We examine concentration regression in future partitions with true identity supplied.

Our contribution is a controlled evaluation of masked self-supervised transfer for quantitative same-system recalibration, with limited labels, held-out identifier partitions, chronological tests and direct competing controls.

PatchTST [18] and multivariate Transformers [19] use masked time-series learning; TS2Vec [20] uses hierarchical contrastive representations. Our compact PatchTST-style regression control compares the complete sensor-aware design with generic masking. Different encoders and capacities prevent isolating sensor identity alone.

PLSR [21] and SVR [22] provide chemometric controls alongside Extra Trees [26] and XGBoost [27]. Farthest-point design is related to concentration-space coverage methods [23]. Nonexchangeable conformal inference [24] and covariate-shift conformal prediction [25] motivate a cautious interpretation of uncertainty under chronological sensor drift.

III. Proposed Method

A. Problem Formulation

For the two raw-response datasets, one gas exposure is represented as the multichannel time series in (1), with the two-gas concentration target in (2)

$$X_i \in R^{T\times S}, (1)$$

where T is the number of temporal samples and S is the number of physical sensors. The target is a two-dimensional concentration vector,

$$y_i=\begin{bmatrix} y_{i1} \\ y_{i2} \end{bmatrix} \in R^2. (2)$$

For the NH3/H2 dataset, $T=64$ and $S=20$. For the dynamic gas mixture dataset, $T=64$ and $S=16$. Pretraining reconstructs responses without concentration labels, emulating an unlabeled mixture pool. The small precisely labeled adaptation set in (3) trains the encoder and regression head under (4).

$$A_K=\{(X_i, y_i)\}_{i=1}^{K}, (3)$$

where K controls the labeled calibration budget. After pretraining, the encoder f_θ and regression head g_ϕ are optimized by

$$(\theta^{¿}, \phi^{¿})=arg\ min_{\theta,\phi} \frac{\sum_{(X_i,y_i)\in A_K} l_{Huber}\left(g_\phi(f_\theta(X_i)), y_i\right)}{¿\,A_K \vee ¿(4)¿}$$

Depending on the adaptation policy, θ is either randomly initialized, frozen after self-supervised pretraining, or updated with a lower learning rate.

B. Shared Temporal Sensor Encoder

The first stage extracts a temporal representation for each sensor. We use the same dilated 1-D convolutional encoder ψ_θ for all physical channels. For sensor s in exposure i,

$$H_{is}=\psi_\theta(X_i[:, s]). (5)$$

The per-sensor temporal encoding in (5) is summarized by combining mean and maximum pooling into the sensor token in (6),

$$h_{is}=\left[MeanPool(H_{is}); MaxPool(H_{is})\right]. (6)$$

Shared weights learn common channel dynamics with fewer sensor-specific parameters; mean and maximum pooling retain overall and strong local responses.

The implemented temporal encoder receives two channels per sensor: the scaled signal and its binary observation mask. It applies three convolutions with channel widths $2\to24\to32\to32$, kernel sizes $(5, 5, 3)$, and dilations $(1, 2, 4)$, respectively. Padding preserves the 64-sample length, and GELU follows each convolution. Mean and maximum pooling produce 64 features, which are projected through a GELU-activated linear layer to a 32-dimensional sensor token.

C. Physical Sensor Identity Encoding

Sharing the temporal network introduces a possible ambiguity. Two different physical sensors may produce temporal features with a similar numerical pattern, even though their selectivity is different. We therefore attach a learned identity embedding e_s to each sensor token,

$$z_{is}=h_{is}+e_s. (7)$$

The jointly learned embedding in (7) preserves physical channel identity without hand-coded gas selectivity.

Sensor identities encode channel correspondence, not one-sensor/one-gas selectivity. Experiments reuse the same array within each dataset and do not establish cross-device transfer.

D. Cross-sensor Self-Attention

The sensor tokens are stacked as in (8)

$$Z_i=¿$$

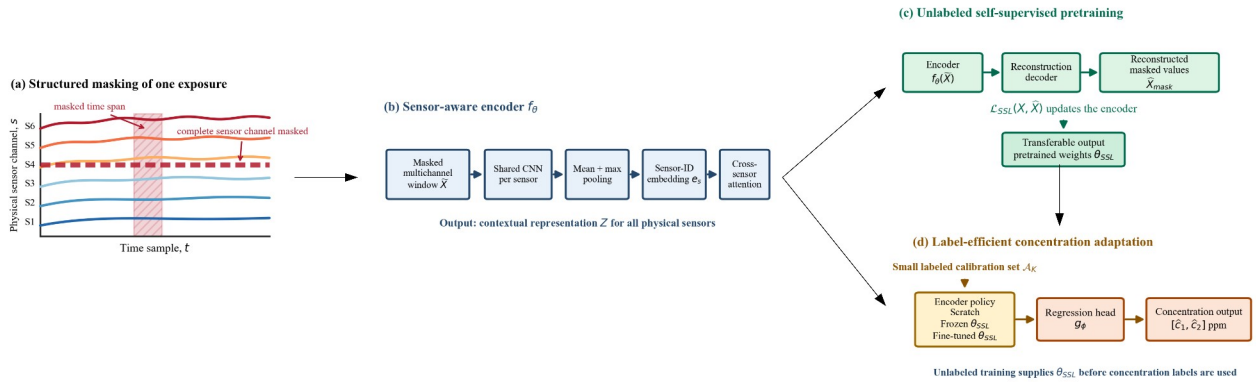


Fig.1. Full sensor-aware pretraining, adaptation pipeline.

We then apply Transformer-style self-attention across the sensor dimension. Query, key, and value matrices are given by (9),

$$Q = Z_i W_Q, K = Z_i W_K, V = Z_i W_V, (9)$$

and the attention output is given by (10)

$$A_i = softmax\left(\frac{Q K^\top}{\sqrt{d_k}}\right) V. (10)$$

Cross-sensor attention models the joint pattern of cross-sensitive responses used to infer gas mixtures.

The implementation uses two pre-normalized Transformer encoder layers. Each layer has a 32-dimensional token size, four attention heads, a 96-unit feed-forward sublayer, GELU activation, and dropout 0.1. A final layer normalization is applied to the contextual sensor tokens. For raw-response regression, the (S) token matrix is flattened; it is not pooled across sensors. The regression head has dimensions (32S), GELU activations, and dropout 0.15 after the 128-unit layer.

E. Structured Masked Self-Supervision

Pretraining reconstructs inputs corrupted by temporal-span and complete-channel masks.

First, contiguous temporal spans are removed. For mask M,

$$M_{bts} = 0 \text{ if } a_k \le t < a_k + w_k \text{ and } B_{bks} = 1, (11)$$

The temporal-span mask in (11) uses a span length sampled as in (12),

$$w_k \sim U\{6, \ldots, 16\}, (12)$$

and per-sensor participation follows (13)

$$B_{bks} \sim Bernoulli(0.45). (13)$$

Here b indexes examples in a minibatch, t the temporal position, s the physical sensor, and k one of three spans sampled independently for each example. Given span width w_k, its start a_k is sampled uniformly from the integers 0 through T − w_k; the example index on a_k and w_k is suppressed. B_bks denotes sensor participation in span k. M is one for observed entries and zero for hidden entries.

We also hide one to three complete sensor channels, requiring reconstruction from the remaining array responses.

The masked reconstruction loss is given by (14):

$$L_{SSL} = \frac{\sum (1-M)(\widehat{X}-X)^2}{max(1, \sum(1-M))} + 0.05\frac{\sum M(\widehat{X}-X)^2}{max(1, \sum M)}. (14)$$

The first term emphasizes the hidden entries. The second provides a smaller consistency penalty on visible values. We also concatenate the mask with the corrupted signal, as in (15),

$$X^{in} = [X \odot M, M], (15)$$

so that a masked value is not confused with a real zero. This avoids the need for target pseudo-labels during pretraining.

Both denominators in (14) are guarded against empty sets. In the present raw-response mask, at most 48 of 64 positions are covered by three spans and at most three of 16 or 20 sensors are fully hidden, so visible entries always remain. The guard does not change this protocol's loss.

F. Adaptation Policies and Calibration-Pair Design

Scratch trains from random initialization; frozen transfer fits only the head; fine-tuning updates the pretrained encoder at a lower learning rate. Fine-tuning is predefined for every gas, without test-based policy selection. Farthest-point design requires known candidate concentrations, unlike label-free pretraining. Its budget includes fitting and validation labels but excludes candidate-pool quantification costs.

Calibration pairs are designed using known concentration coordinates by farthest-point sampling, following the limited-experiment calibration motivation in [14]. Let P be the candidate set and S_k the selected set after k steps. The next pair is

$$p_{k+1} = arg \max_{p \in P \setminus S_k} \min_{q \in S_k} \| \tilde{p} - \tilde{q} \|_2, (16)$$

where $\tilde{p}$ denotes the normalized concentration pair.

IV. Datasets and Experiment Protocol

A. NH3/H2 dataset

The NH3/H2 dataset [29] contains 162,404 observations from 20 Nano MEMS sensor channels, grouped into 378 recorded sequences, with concentrations of 0–500 ppm for each gas. Acquisition took place over six days. The six partitions of 63 sequences were constructed from consecutive Combination identifiers and were not designed as day splits. Day-by-day mixture details were not logged, so a reliable exposure-to-day mapping and leave-one-day-out evaluation cannot be reconstructed. Training and test partitions may share acquisition days; these folds do not establish between-day generalization.

The primary NH3/H2 evaluation holds out each of the six partitions in turn, uses the preceding cyclic partition for adaptation, and uses the remaining four for pretraining. Because some pretraining partitions follow the test partition, this is held-partition validation, not prospective forecasting. The secondary fixed split uses partitions 1–4 for pretraining (252 exposures; 1,260 windows), partition 5 for adaptation, and partition 6 for testing. Five 64-sample training windows use offsets 0–4; tests use offset zero only. Related image-based mixed-gas estimation is described by Ha et al. [30].

Within every fold, complete exposures are assigned before window extraction. Response scaling is fitted exclusively to that fold's pretraining responses. Adaptation fitting and validation exposures are disjoint, and all policies use identical group assignments within each seed and budget. Test exposures are excluded from preprocessing fits, pretraining, calibration design, fitting, and checkpoint selection. A budget counts both fitting and validation concentration pairs.

B. Dynamic Gas Mixture Dataset

The dynamic gas mixture dataset is publicly available from the UCI Machine Learning Repository [15]. Its acquisition and continuous-monitoring protocol are described in [8]. This work uses the CO and ethylene portion, prepared as 245 chronological exposure runs with 16 response channels and 1,225 stored windows. Each run contributes five 64-sample windows at offsets 0, 20, 40, 60, and 80. The processed target scale is 0–533.33 ppm for CO and 0–20 ppm for ethylene.

Primary CO/ethylene validation divides the 245 acquisition-ordered groups into seven blocks and evaluates three rolling folds, each with three pretraining blocks followed by one adaptation block and one future test block. Secondary fixed-split experiments retain five blocks of 49 groups: blocks 1–3 pretrain, block 4 adapts, and block 5 tests. Budgets are 5, 10, 20,

and 24 unique concentration pairs. Training uses five stored offsets; the test uses one window per group.

C. Sensor Drift Dataset

To evaluate long-term drift, we use the public sensor drift dataset [16]. The drift-compensation setting is described by Vergara et al. [17], and calibration with a limited number of experiments is examined in [14]. The dataset contains 13,910 complete records organized into ten time-based partitions spanning months 1–36. Each record holds 128 values: eight engineered descriptors for each of 16 sensor channels, so that each sample is a 16×8 matrix. The eight descriptors per sensor are two resistance-change features, three rising-response exponential-moving-average features, and three decaying-response exponential-moving-average features; they are engineered descriptors rather than consecutive temporal samples. The six gases are ethanol, ethylene, ammonia, acetaldehyde, acetone, and toluene.

Partitions 1–6 (months 1–20) pretrain the descriptor encoder; partition 7 (month 21) provides adaptation labels; partitions 8–10 are separate future tests. We evaluate 1, 2, 5, 10, and 20 labels per gas with 10 seeds. At 20 labels/gas, 16 fit and four validate each gas, identically across policies. One label/gas has no disjoint holdout and therefore uses the predefined final epoch; larger budgets select checkpoints using adaptation-only validation.

Raw temporal curves are unavailable in this benchmark, so the self-supervised masking is adapted to the representation. A complete sensor is masked independently with probability 0.18. One descriptor group—the resistance pair, the rising descriptors, or the decaying descriptors—is selected per record and hidden independently on each sensor with probability 0.35, and individual descriptors are additionally masked with probability 0.08. The shared principle remains structured reconstruction of physically grouped sensor information. The descriptor encoder maps the eight scaled descriptors and eight mask indicators through a shared 16→64→32 GELU multilayer perceptron, adds a learned 32-dimensional sensor-identity embedding, applies the same two-layer four-head Transformer, and pools mean and maximum across the 16 contextual tokens followed by a 64-unit GELU layer with dropout 0.15.

The supervised objective for this extension contains only the gas-conditional concentration term. Six gas-conditional regression heads output normalized log-concentration values, and the known gas identity selects the corresponding output; no gas-classification head is trained or evaluated. The reported concentration MAE is therefore an oracle-class conditional metric: it assumes gas identity is available and does not represent end-to-end concentration prediction for an unknown gas. The predefined 20-label analysis uses the normalized log targets in (17); fixed-scale log and linear losses are additional sensitivity controls.

Figure 2 preserves the original secondary fixed-split schematic and the primary chronological drift roles. Primary NH3/H2 cyclic folds and CO/ethylene rolling folds are specified above. Pretraining withholds concentration labels; adaptation and test roles remain disjoint.

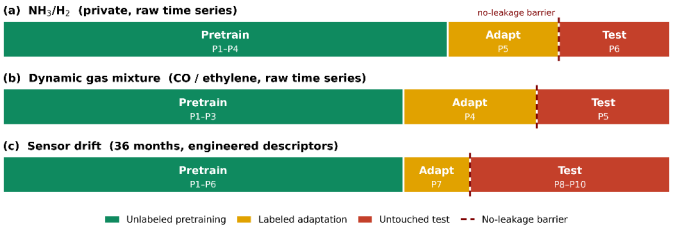


Fig. 2. Leakage-controlled secondary fixed splits and chronological drift roles; the primary multi-fold protocols are specified in Section IV.

D. Preprocessing and Optimization

For the two raw-response datasets, a per-sensor robust scaler is fitted on unlabeled pretraining windows only. The center is the median and the scale is the interquartile range; scaled values are clipped to $[-12, 12]$. Targets are divided by fixed implementation scales before Huber training: $[500, 500]$ ppm for the NH3/H2 dataset and $[533.33, 20]$ ppm for the dynamic gas mixture dataset. Predictions are transformed back to ppm and clipped to these ranges.

Raw-response pretraining uses AdamW for 60 epochs with learning rate 8×10^{-4}, weight decay 10^{-4}, batch size 64, and gradient-norm clipping at 2. Supervised fitting uses 120 epochs and Huber loss with $\delta = 0.1$ in normalized target space. Scratch training uses learning rate 5×10^{-4}. Frozen transfer trains only the head at 10^{-3}. Fine-tuning uses 8×10^{-5} for the encoder and 8×10^{-4} for the head. All supervised policies use weight decay 2×10^{-4} and batches of at most 64 samples.

Sensor drift dataset features are robust-scaled using partitions 1–6 with the 10th–90th percentile range and clipped to $[-10, 10]$. Descriptor pretraining uses 40 AdamW epochs, learning rate 8×10^{-4}, weight decay 10^{-4}, batch size 256, and gradient clipping at 2. Supervised sensor drift dataset fitting uses 120 epochs, weight decay 2×10^{-4}, and adaptation-only minimum-loss checkpoint selection, except for the one-label final-epoch protocol. The head learning rate is 8×10^{-4}; the encoder rate is 5×10^{-4} from scratch and 8×10^{-5} during fine-tuning. Concentration is transformed per gas as

$$u_i = \frac{\log(1 + c_i) - \mu_{r_i}}{\sigma_{r_i}}, \quad (17)$$

where r_i is the true gas class and μ_r, σ_r are estimated from the supervised fitting subset only. The sensor drift dataset Huber term uses $\delta = 0.5$.

All revised experiments use seeds 42, 142, 242, 342, 442, 542, 642, 742, 842, and 942, with seed-specific pretraining. Approximately 20% of selected exposure groups are reserved for validation. Random and farthest-point calibration designs are crossed with scratch and fine-tuning at every fixed-split budget. Full multi-fold coverage uses all available adaptation pairs, whose number can differ between rolling folds.

The NH3/H2 array uses 20 MEMS SnO2 channels with Pt/Ag modification, as described in Supplementary "NH3/H2 dataset and acquisition details." Temperature and relative humidity were neither controlled nor measured during acquisition, so their variation and contribution to response differences cannot be quantified retrospectively. Digital audits

found no identical stored signal fingerprints or identifier overlap across 11 split definitions; they do not establish session independence. The 0–500 ppm range does not validate low-concentration ambient monitoring.

Each NH3/H2 sequence starts at gas delivery and contains 180 s of exposure followed by 700 s of recovery. The configured sampling interval is 2.0 s. The predefined 64-sample early window covers about 128 s nominally (approximately 131 s from aggregate sample counts), or roughly 71–73% of exposure. Middle, late and full-cycle sensitivity inputs also have 64 samples, with scalers refitted on their pretraining windows. Full-cycle resampling includes recovery, rather than only the 180-s gas exposure. Exact index definitions and the approximate timing calculation are supplied in the supplement.

The generic SSL control uses eight nonoverlapping length-8 patches per channel, 32-dimensional tokens, two four-head Transformer layers, 96-unit feed-forward blocks, shared channel weights, temporal positions, and three masked patches per channel. It has no physical sensor embeddings or cross-channel attention. Temporal mean pooling feeds the same concatenated-channel regression head, so channel ordering remains available. Instance normalization is omitted to preserve absolute responses and match preprocessing; capacity differs and is reported. Pretraining/adaptation data, budgets, seeds, scaling and checkpoint rules match the proposed model.

Classical regressors receive flattened pretraining-scaled windows: 1,280 NH3/H2 or 1,024 CO/ethylene features. PLSR and separate-target RBF SVR additionally standardize features using fitting data only. Hyperparameters are selected by adaptation-validation normalized MAE. Tree settings are fixed. At 20 drift labels/gas, historical log-Huber is compared with log1p(c)/log1p(1000), linear c/1000 Huber, and linear c/1000 MAE. Below 20 labels/gas the fixed log scale avoids estimating a standard deviation from tiny fitting sets. The 1000-ppm divisor is a numerical unit, not an asserted physical maximum. Constant controls use only the per-gas fitting mean.

Representative seed-42 pretraining reruns on one NVIDIA GeForce RTX 3060 took 38.8 s for NH3/H2 (1,260 windows) and 19.9 s for CO/ethylene (735 windows), each for 60 epochs with batch size 64. Timings include model initialization and optimization but exclude data loading, scaling and checkpoint writes. They are per-model measurements, not reconstructed total campaign costs; checkpoints are reused across adaptation budgets. Software and timing details are supplied in the supplement.

E. Evaluation Metrics

Besides using standard metrics like MAE, RMSE and R^2, we also measure the 90th percentile of absolute error , given by (18) to examine large errors.

$$Q_{0.9,\,j} = quantile_{0.9}¿$$

Relative transfer gain is defined as in (19):

$$Gain(\%) = 100\,\frac{E_{scratch} - E_{transfer}}{E_{scratch}}, \quad (19)$$

For sensor drift dataset concentration regression, we report the unweighted mean across six gases in (20),

$$MacroMAE = \frac{1}{6}\sum_{r=1}^{6} MAE_{r}. \quad (20)$$

We report seed means and sample SD (ddof=1). Paired 95% bootstrap intervals use 5,000 crossed seed-and-fold resamples, weighting folds equally and conditioning on observed test exposures. Repeated predictions are not independent exposures, and overlapping training folds limit population inference. Fixed-split controls resample paired seeds; the control comparisons are exploratory and not multiplicity-adjusted. Drift NMAE divides each gas MAE by its concentration range in historical partitions 1–6, whose labels are used for reporting only. This permits an equal-gas macro-NMAE when a test-partition range is zero; the test-range diagnostic is also retained.

Primary intervals in Table I are unadjusted. As a post-review multiplicity sensitivity check, we also report 98.75% marginal percentile intervals for a four-comparison Bonferroni family, using 200,000 crossed seed/fold draws. A second random-number seed checks numerical stability. This adjustment does not remedy dependent training folds or missing acquisition-day labels (Supplementary Table S9).

V. Results

A. Self-Supervised Pretraining

Figure 3 shows ten-seed reconstruction diagnostics. Losses across differently scaled datasets are not concentration-accuracy comparisons; subsequent results use held-out predictions.

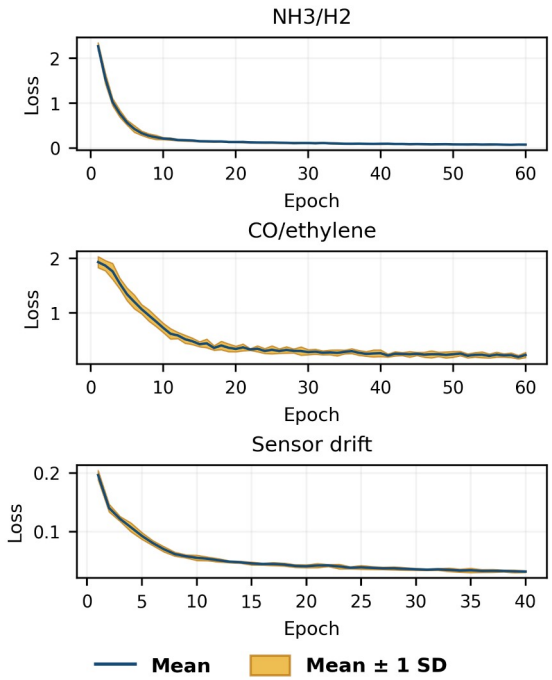


Fig. 3. Reconstruction loss over 10 seeds: blue line, mean; orange band, mean ± one seed SD. Raw-response pretraining uses 60 epochs; descriptor pretraining uses 40.

B. Primary Multi-Fold Validation

Table I compares fine-tuning with scratch under matched adaptation groups and holdouts. The four unadjusted paired 95% intervals exclude zero. In a four-comparison Bonferroni sensitivity analysis, the NH3 interval includes zero; the other three remain positive (Supplementary Table S9). NH3 is therefore the weakest primary case. CO/ethylene intervals are indicative because they resample only three chronological folds. Inference is conditional on observed exposures, not independent acquisition campaigns.

Table I. Primary Multi-Fold MAE and Paired Difference (ppm); Unadjusted 95% Intervals

Gas	Scratch	Fine-tuned	Difference	95% interval
NH3	31.01	26.69	4.31	[0.38, 8.30]
H2	49.20	38.63	10.58	[3.57, 18.78]
CO	89.95	65.22	24.74	[12.85, 37.53]
Ethylene	2.611	1.939	0.672	[0.247, 1.143]

Relative reductions are NH3: 13.9%, H2: 21.5%, CO: 27.5%, Ethylene: 25.7%. Across-fold SD of fine-tuned mean MAE is NH3: 11.07 ppm, H2: 10.59 ppm, CO: 14.71 ppm, Ethylene: 0.448 ppm. NH3/H2 includes 378 distinct test identifiers over six folds (3,780 predictions per gas across seeds); CO/ethylene includes 105 identifiers over three rolling tests (1,050 predictions per gas). Supplementary Table S1 retains every fold and all policies.

Primary improvements do not erode systematically across ten seeds. Supplementary Table S2 and Fig. S1 report seed prefixes and random-order sensitivity; all ten seeds are retained.

Transfer reverses in identifier fold 4: NH3 MAE increases from 16.22 to 18.83 ppm and H2 from 25.70 to 26.36 ppm. Aggregate gains therefore do not imply benefits in every partition. The available records do not support attributing this reversal to a particular day or environmental condition.

C. Fixed-Split Label-Budget Analysis

Table II reports all adaptation policies as mean ± seed SD. Budgets include validation labels. These fixed-split experiments are secondary to the multi-fold evaluation; neither the policy nor the budget is chosen from test performance. At full budget the paired means favor fine-tuning over scratch, but errors do not decrease monotonically at every budget.

Table II. Fixed-Split MAE (ppm), Farthest-Point Calibration-Pair Design, Mean ± SD Over Ten Seeds

Gas	Pairs	Scratch	Frozen	Fine-tuned
NH3	5	86.39 ± 36.11	82.85 ± 31.89	86.02 ± 34.16
NH3	10	67.52 ± 40.50	41.08 ± 7.01	38.92 ± 4.56
NH3	20	36.84 ± 7.89	31.10 ± 2.80	29.53 ± 2.38
NH3	40	29.86 ± 2.74	25.09 ± 2.01	25.22 ± 2.10
NH3	57	21.59 ± 3.25	19.26 ± 1.76	17.36 ± 1.54
H2	5	161.51 ± 22.40	129.17 ± 12.17	128.91 ± 16.84
H2	10	124.16 ± 28.13	78.73 ± 18.12	81.88 ± 19.54
H2	20	70.73 ± 30.45	48.04 ± 8.65	48.38 ± 9.70
H2	40	51.53 ± 6.60	41.26 ± 2.34	41.67 ± 2.75
H2	57	41.20 ± 4.27	34.77 ± 2.70	31.60 ± 2.29
CO	5	163.06 ± 47.96	125.10 ± 34.73	111.46 ± 23.96
CO	10	156.04 ± 27.60	77.08 ± 25.00	80.99 ± 27.98
CO	20	89.30 ± 39.24	49.96 ± 8.66	47.42 ± 7.79
CO	24	63.63 ± 10.68	46.81 ± 9.60	47.45 ± 7.38
Ethylene	5	5.226 ± 1.469	3.916 ± 1.680	3.468 ± 1.248
Ethylene	10	3.610 ± 1.421	2.449 ± 0.464	2.418 ± 0.484
Ethylene	20	2.465 ± 0.745	1.879 ± 0.534	1.843 ± 0.506
Ethylene	24	2.149 ± 0.359	1.668 ± 0.241	1.627 ± 0.275

For example, fine-tuned CO MAE is 47.42 ± 7.79 ppm at 20 pairs and 47.45 ± 7.38 ppm at 24 pairs. Figure 4 retains all three policies rather than a test-selected policy summary.

Fixed-split gains attenuate with additional seeds, especially NH3 (32.4% for the first three versus 19.6% for all ten). Detailed seed sensitivity is supplementary; primary conclusions use multi-fold evidence.

Table IIa. Label-Budget Comparison, Farthest-Point Fixed Split; MAE (ppm), Mean ± SD

Gas	Transfer: 10 pairs	Scratch: 20 pairs
NH3	38.92 ± 4.56	36.84 ± 7.89
H2	81.88 ± 19.54	70.73 ± 30.45
CO	80.99 ± 27.98	89.30 ± 39.24
Ethylene	2.418 ± 0.484	2.465 ± 0.745

Table IIa expresses label efficiency in labeled two-gas concentration pairs, including fitting and validation. At 10 pairs, transfer approximately matches scratch at 20 for NH3 and ethylene and has lower mean CO error, but higher H2 error. These descriptive comparisons are not equivalence tests or a universal 50% label-saving guarantee. Pairs, exposure records and overlapping windows are distinct units.

At five pairs, NH3 shows essentially no mean benefit (86.39 ± 36.11 to 86.02 ± 34.16 ppm). An error near 86 ppm is 17.2% of its 0–500 ppm range and does not demonstrate useful precise quantitative calibration. Sparse calibration coverage and training variability are possible contributors, not established causes. This is consistent with NH3's weak primary and component-ablation evidence.

D. Calibration-Design Control

Calibration-design effects depend on both target and budget (Supplementary Table S3). At five pairs, farthest-point design reduces fine-tuned CO MAE relative to random by 31.61 ppm (unadjusted 95% interval 13.34–51.92). At 40 pairs, random design instead has lower MAE: random minus farthest-point differences are −4.46 ppm for NH3 and −8.32 ppm for H2 under scratch training, and −3.37 and −4.89 ppm under fine-tuning; all four exploratory intervals exclude zero. These are known-coordinate calibration designs, not active acquisition of unknown labels. Full-budget designs coincide when all candidate pairs are used.

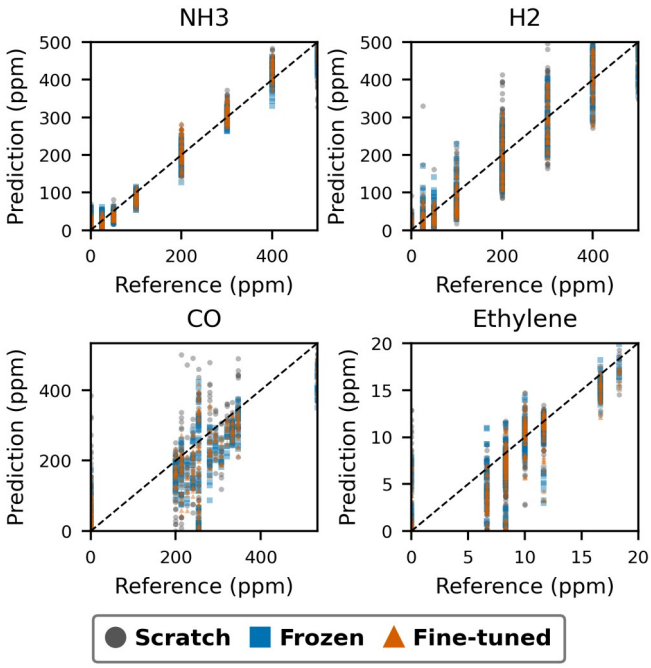


Fig. 4. Secondary full-budget predictions from all 10 seeds: gray circles, Scratch; blue squares, Frozen; orange triangles, Fine-tuned. Points repeat test exposures across seeds; the dashed line is identity, not a fitted trend.

Table III. Five-Pair Design Control, MAE (ppm), Mean ± SD

Gas	Design	Scratch	Fine-tuned
NH3	Random	98.51 ± 29.07	89.75 ± 29.49
NH3	Farthest	86.39 ± 36.11	86.02 ± 34.16
H2	Random	159.51 ± 12.93	143.53 ± 30.16
H2	Farthest	161.51 ± 22.40	128.91 ± 16.84
CO	Random	179.22 ± 31.43	143.07 ± 22.16
CO	Farthest	163.06 ± 47.96	111.46 ± 23.96
Ethylene	Random	4.890 ± 1.836	4.323 ± 1.682
Ethylene	Farthest	5.226 ± 1.469	3.468 ± 1.248

E. Baseline Comparison

The proposed model has the lowest mean MAE on none of the four targets at full budget (Table IV). Generic SSL leads on NH3 (16.13 versus 17.36 ppm), SVR on H2 (31.37 versus 31.60 ppm), Extra Trees on ethylene (1.398 versus 1.627 ppm), and PLSR on CO (25.57 versus 47.45 ppm). Proposed CO MAE is 1.86 times PLSR MAE despite improvement over scratch. A lower-complexity latent linear model may better suit the small calibration set and this response representation; this is a possible explanation, not an identified mechanism. Close mean rankings are not equivalence tests, and overlapping SDs do not establish statistical ties. Tuning grids and all-budget results are supplied in the supplement.

Paired full-budget generic-SSL comparisons favor the sensor-aware model on H2, favor generic SSL on NH3 and ethylene, and are inconclusive for CO. Lower-budget effects are also mixed (Supplementary Table S4). Encoder and capacity differences preclude attributing these contrasts solely to sensor identity.

Table IV. Full-Budget Baseline MAE (ppm), Mean ± SD

Model	NH3	H2	CO	Ethylene
Scratch	21.59 ± 3.25	41.20 ± 4.27	63.63 ± 10.68	2.149 ± 0.359
Extra Trees	21.63 ± 1.73	37.33 ± 2.04	90.46 ± 7.17	1.398 ± 0.197
XGBoost	26.96 ± 4.61	48.65 ± 2.38	55.90 ± 11.13	2.491 ± 0.352
PLSR	28.44 ± 4.76	75.50 ± 18.25	25.57 ± 8.48	1.994 ± 0.377
SVR	20.74 ± 1.62	31.37 ± 4.53	53.01 ± 7.26	2.013 ± 0.226
Generic SSL	16.13 ± 1.23	35.18 ± 2.74	52.52 ± 12.59	1.437 ± 0.275
Fine-tuned	17.36 ± 1.54	31.60 ± 2.29	47.45 ± 7.38	1.627 ± 0.275

F. Ablation and Window Sensitivity

Table V. Full-Budget Ablation Minus Full-Model MAE (ppm), Mean ± SD

Removed	NH3	H2	CO	Ethylene
No sensor ID	-0.06 ± 1.66	4.73 ± 3.73	5.17 ± 10.05	0.097 ± 0.462
No attention	-3.01 ± 1.49	1.99 ± 3.50	10.27 ± 12.18	-0.140 ± 0.264
No time mask	2.10 ± 2.17	1.73 ± 3.30	7.93 ± 14.86	-0.016 ± 0.360
No sensor mask	-0.78 ± 2.00	4.34 ± 3.75	1.83 ± 8.34	0.027 ± 0.215

Table V reports within-seed ablation-minus-full-model differences: positive values favor the complete model. Effects vary by gas; sampling and optimization cannot be separated from physical selectivity, and no component is uniformly beneficial.

Removing sensor identity changes NH3 MAE by -0.06 ± 1.66 ppm, compared with 4.73 ± 3.73 ppm for H2. Removing attention lowers NH3 error by 3.01 ppm on average. Thus the original large cross-gas identity-effect ratio is not treated as a stable physical constant, and the attention module is not uniformly beneficial.

CO ablations are variable: removing attention gives 10.27 ± 12.18 ppm and removing temporal masking 7.93 ± 14.86 ppm. Exploratory paired-seed 95% bootstrap intervals are [2.68, 16.94] and [−0.33, 17.00] ppm, respectively (Supplementary Table S10). Temporal masking is compatible with no effect; attention has a positive conditional mean contrast, but its large variability limits a robust component-benefit claim. Neither SD overlap nor these unadjusted intervals establishes a general physical mechanism.

Table VI. Window Sensitivity, MAE (ppm), Mean ± SD

Window	NH3 scratch	NH3 transfer	H2 scratch	H2 transfer
Early	21.59 ± 3.25	17.36 ± 1.54	41.20 ± 4.27	31.60 ± 2.29
Middle	34.55 ± 5.01	29.60 ± 3.23	33.99 ± 4.10	35.60 ± 3.08
Late	56.71 ± 5.88	54.75 ± 4.40	52.12 ± 7.87	51.28 ± 4.89
Full-cycle	20.24 ± 1.76	18.05 ± 2.48	38.36 ± 2.56	35.75 ± 2.46

Window placement materially changes both absolute error and transfer effects (Table VI). Fixed-split early-window H2 improves from 41.20 to 31.60 ppm (23.3%); middle-window transfer reverses the benefit (33.99 to 35.60 ppm, −4.7%); full-cycle resampling yields 38.36 to 35.75 ppm (6.8%). The 21.5% headline instead comes from primary multi-fold evaluation using the same predefined early-window definition. It should not be generalized to other window placements. Middle/late windows sample the recovery part of the recorded cycle; full-

cycle resampling spans exposure and recovery. No window is selected using test performance.

G. Long-Term Drift

The primary 20-label/gas transform is historical per-gas log-Huber; the constant predictor uses the same fitting records. Scratch minus fine-tuned macro-MAE is partition 8: 4.50 ppm (95% interval 2.55 to 6.19); partition 9: 4.16 ppm (95% interval 2.41 to 6.22); partition 10: -0.71 ppm (95% interval -8.40 to 6.91). These intervals resample seeds, conditional on each test partition. The comparison describes concentration regression with true gas identity supplied; it excludes classification errors.

Table VII. Oracle-Class Drift Macro-MAE (ppm), Mean ± SD

Policy	Partition 8	Partition 9	Partition 10
Constant	52.91 ± 3.07	46.13 ± 3.07	181.05 ± 2.13
Scratch	23.95 ± 3.46	22.68 ± 2.86	142.34 ± 11.06
Frozen	19.09 ± 2.25	18.91 ± 3.25	137.43 ± 13.49
Fine-tuned	19.45 ± 3.37	18.52 ± 4.60	143.05 ± 12.49

Partition 10 shows no clear improvement: its paired interval includes zero. Constant and normalized-error controls remain necessary; favorable conditional comparisons do not establish severe-drift deployment robustness.

Table VIII. Fine-Tuned Per-Gas NMAE (% of Historical Range), Mean ± SD

Gas	Partition 8	Partition 9	Partition 10
Ethanol	0.49 ± 0.52	0.40 ± 0.45	10.76 ± 0.90
Ethylene	5.04 ± 1.40	4.23 ± 1.50	21.72 ± 3.51
Ammonia	3.34 ± 0.93	1.74 ± 1.02	36.00 ± 3.06
Acetaldehyde	5.65 ± 2.43	10.02 ± 2.92	24.19 ± 2.34
Acetone	6.81 ± 0.33	4.55 ± 1.27	61.11 ± 12.72
Toluene	18.76 ± 8.32	30.16 ± 11.61	22.61 ± 4.58
Macro	6.68 ± 1.74	8.52 ± 2.48	29.40 ± 3.07

Table VIII uses each gas's fixed historical concentration range and weights gases equally; it avoids allowing high-ppm analytes to dominate the normalized macro score. All-policy per-gas MAE and NMAE, reference ranges, and all five budgets are in Supplementary Tables S5–S6. A one-label/gas run is a fixed-epoch tiny-sample test, not a validated checkpoint comparison.

Under fixed-log scaling, fine-tuned one-label/gas macro-MAE is 8: 40.73 ± 24.75 ppm, 9: 39.47 ± 20.58 ppm, 10: 206.22 ± 24.05 ppm. Its large seed variation and the substantial partition-10 error limit claims of extreme label efficiency. Budget effects are not uniformly monotonic. The fixed-log 20-label control is retained to separate budget effects from the primary protocol's change to fitted-log scaling at 20 labels.

Table IX. Fine-Tuned Loss/Scaling Sensitivity, Macro-MAE (ppm), Mean ± SD

Target / loss	Partition 8	Partition 9	Partition 10
Log fitted SD	19.45 ± 3.37	18.52 ± 4.60	143.05 ± 12.49
Log fixed	22.68 ± 5.59	26.11 ± 6.19	184.96 ± 9.99
Linear Huber	26.94 ± 4.16	25.50 ± 6.16	145.56 ± 11.05
Linear MAE	22.48 ± 4.17	20.70 ± 4.75	137.55 ± 12.87

Log-Huber and linear MAE optimize different error geometries. Table IX tests alternatives without choosing the primary transform from test scores. Fixed scaling avoids fitting log variance from 16 records per gas. Supplementary Table S7 retains every policy and fitted scale.

Historical fitted-log Huber has the lowest observed fine-tuned macro-MAE among these transforms in partitions 8 and 9. In partition 10, linear MAE gives 137.55 ± 12.87 ppm versus 143.05 ± 12.49 ppm for fitted-log Huber, while fixed-log Huber rises to 184.96 ± 9.99 ppm. The apparent severe-drift benefit of changing the loss is therefore not a uniform scaling improvement, and the linear-MAE transfer-versus-scratch interval also includes zero.

H. Tail Error, Capacity, and Uncertainty

Table X. Full-Budget Mean R^2, RMSE and Q0.9 Across Ten Seeds

Gas	Policy	R^2	RMSE	Q0.9
NH3	Scratch	0.962	31.16	48.79
NH3	Frozen	0.975	25.57	44.64
NH3	Fine-tuned	0.979	23.29	38.11
H2	Scratch	0.881	57.78	96.54
H2	Frozen	0.925	46.01	78.41
H2	Fine-tuned	0.935	42.90	74.15
CO	Scratch	0.617	94.89	162.79
CO	Frozen	0.771	73.58	110.76
CO	Fine-tuned	0.777	72.93	109.51
Ethylene	Scratch	0.558	3.257	5.769
Ethylene	Frozen	0.749	2.457	3.703
Ethylene	Fine-tuned	0.742	2.483	4.130

CO remains quantitatively limited even after transfer: mean R^2 is 0.777, and Q0.9 is 109.51 ppm on the 0–533.33 ppm scale. Relative improvement must not be equated with adequate quantitative calibration. Table X includes scratch tail errors and frozen transfer as well as fine-tuning.

NH3/H2: 117,514 proposed parameters (31,272 encoder; 86,242 head), all trainable in fine-tuning versus 86,242 in frozen transfer; generic SSL has 108,098. The seed-42 full-budget fitting subset has 46 distinct exposures, giving 2,555 proposed parameters per fitting exposure. CO/ethylene: 101,002 proposed parameters (31,144 encoder; 69,858 head), all trainable in fine-tuning versus 69,858 in frozen transfer; generic SSL has 91,714. The seed-42 full-budget fitting subset has 19 distinct exposures, giving 5,316 proposed parameters per fitting exposure. Five overlapping windows do not multiply the independent sample count. Supplementary Table S8 gives low/full-budget model sizes and drift counts; high parameter-to-exposure ratios remain an overfitting concern despite regularization and validation.

Chronological/session shift violates the exchangeability assumption of standard split-conformal coverage [24], [25]. Such intervals are diagnostic under shift, not deployment guarantees. Shift-aware uncertainty remains unevaluated.

VI. Discussion and Conclusion

Masked self-supervised pretraining reduces mean error against scratch in this same-system calibration evaluation. Under predefined fine-tuning, primary reductions are 13.9% for NH3, 21.5% for H2, 27.5% for CO and 25.7% for ethylene; NH3/H2 uses the early-response window. Fixed-split pair-count comparisons show approximately comparable NH3/ethylene

errors and lower mean CO error with 10 transfer pairs versus 20 scratch pairs, but not an H2 match at those budgets.

The proposed model is not the best full-budget baseline on any of the four targets; CO MAE is 1.86 times that of PLSR. Sensor identity and attention are not uniformly beneficial, particularly for NH3. NH3 also has negligible five-pair improvement and is the weakest multiplicity-adjusted primary case. Window and calibration-design effects vary with target and budget. Remaining absolute and tail errors limit quantitative usefulness.

Oracle-class drift regression improves in partitions 8-9 but not clearly in partition 10. The evaluation assumes known gas identity. Constant, normalized-error and loss controls show that representation reuse alone does not establish severe-drift robustness.

Ten seeds quantify training/design variability, not independent acquisitions. NH3's four-comparison adjusted interval includes zero; CO/ethylene intervals remain indicative with only three folds. NH3/H2 acquisition spanned six days, but missing day-by-day mixture logs prevent reliable day-held-out evaluation. Temperature and humidity were neither controlled nor measured. Consequently these identifier folds cannot establish between-day or environmental robustness. Cross-device transfer remains unevaluated.

Label withholding tests the learning premise retrospectively, not acquisition-cost savings. Known-coordinate calibration design excludes candidate-pool quantification costs. Future experiments will measure and record environmental conditions; exposure-linked day/session logs are needed for independent-day evaluation. Prospective unquantified-mixture pretraining, acquisition-cost accounting, and shift-aware uncertainty remain future work.

VII. Data and Code Availability

The NH3/H2 dataset is identified by Harvard Dataverse DOI 10.7910/DVN/LWUSKO [29]; the public CO/ethylene and drift datasets are cited in [15] and [16]. Study code is linked at https://github.com/ngdvan/Transfer-learning-Enose. The accompanying revision package contains the analysis scripts, split-audit records, unrounded result exports and representative compute measurements used here. Exposure-to-day mixture logs and environmental measurements were not recorded and are not available.

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